Testnet findings: the 23 Oct 2026 recording (season 224 round 3 to season 225)

What was recorded. `testnet/recordings/testnet_s224_s225.jsonl` has 44,296 frames, recorded from 10:47 to 11:50 EDT on `cornelltradingcompetition.org`. It covers:

- season 224 round 3 from real 218.3 s (storm age 160; the brief's "real 160 s" is the storm age);
- all of season 225 (Pia, Rex and Sia);
- about 50 s of season 226.

Where the numbers come from.

- Four analysis reports (`feed/`, `books/`, `events/`, `rebacktest/`) were checked by an independent verifier (`verify/`). This document keeps only claims the verifier confirmed or corrected.
- Where the verifier did not re-test a claim, the writer re-ran or re-measured it (`writer/`). Anything that rests on neither is labelled, and **[I]** marks an inference or an untested rule.
- Every number cites its output file:
  - `vNN` is `verify/vNN_*.out`;
  - `wNN` is `writer/wNN_*.out`, with a section letter where the file has sections;
  - `w00 engine`, `w00 snapshot` and `w00 doublecount` are the writer's re-runs of the feed report's `engine_sens.py`, `snapshot_resync_check.py` and `doublecount_check.py`. The outputs are identical to the published ones apart from timing lines.
- Re-run everything with `sh writer/run_all.sh <recording> <SP>`.

How much to trust it. This is one season of book evidence: 3 full rounds plus one partial, 3 formations and 8 damage reports. The recorder never traded, so our own fills and impact are not observed. Single-round results are anecdotes.

Key findings

1. **The testnet book is not the research consensus plus a spread.**
   - Season 225 opened at the packet's starting prices: 8 of 12 symbols were within 25 bp of the packet, against 6 of 12 within 25 bp of `market.csv` (`v09`).
   - In 225 r1, HAVEN's book sat about 13% above the consensus from 300 to 540 s (56.0-56.3 against 49.4-49.7). It snapped to 52.75 at 552 s (`v10`).
   - Other participants' net flow ran *against* that gap (correlation −0.69), so the gap is the house's own quote level, not skew from flow (`v09`).
   - In 225 r3 and 224 r3 the book tracked the consensus: the RMS gap was 27-165 bp and 22-112 bp, against 173-1,160 bp in 225 r1 (`v10`).
   - **Action:** price every edge off the book. P&L measured on the consensus cannot be checked against fills.
2. **Costs: small clips are cheaper than the RT model, large clips much dearer, and 72-120 s is the expensive window.**
   - Storm half-spread is a median 12-23 bp. An L/4 clip costs a median 34-51 bp for stocks and 16 bp for ETFs (`v03`).
   - Fitting the RT form gives h = 11-16 bp and γ = 0.8-1.3% per full limit for stocks, against the model's 0.25% and 0.5% (`v03`).
   - Every book uses the same ladder: about 0.10 / 0.07 / 0.18-0.30 / 0.24 L, then a 3 L wall about 8% from mid (`w01 D`). A full-limit stock clip therefore costs 304-328 bp (`v03`).
   - Clips cost 2-7× the storm level in 72-120 s (`v03`).
3. **The formation jump shows on the books, but late, so H1 with the playbook's 72-78 s exit made about $0.**

- The book move has the consensus sign in 29 of 33 symbol-rounds, at a median 1.20× its size (`v04`).
- The book is half done at a median 72.0 s for stocks and 77.3 s for ETFs (`w06`). The consensus reprices at 64 s and 70 s.
- Full-size H1 made **+$65 on the books**, against +$7,389 for the same capped trades on the consensus with the RT model (`v04`).

4. **S1-r3 (the settlement trade after the round-3 final report) is confirmed, and the window is longer on the books.**
   - For stock moves of at least 1%, the book has done 20% of its move at +4 s and 91% at +8 s. The consensus has done 99% by +4 s (`v05`).
   - A sweeper starting 1 s after the report finds at least $924 (224 r3) and $927 (225 r3) (`v05`).
   - The 746-s sanity check would cancel PORT, HAVEN and ARCH, the profitable names, in both rounds (`v05`).
5. **S3 (the core engine taker) is not yet proven on the books.**
   - S3 alone in season 225 made –$3,790 (`r11`, reproduced in `v07`) and –$2,175 (`v12`) on the books, against +$3,569 and +$3,000 on the consensus.
   - Round 2 drives the loss (books –9,407): the engine was also wrong there on the consensus. Round 3 made +7,738 on the books (`v12`).
   - The S3 signal is not reversed on the books (`v08`).
6. **Re-backtest with measured costs, still priced on the consensus.**
   - The Plan as written earns **10,713 [9,811]**, with p5 –7,821 (`v06`).
   - The re-backtest's recommended configuration, with H1 at full size, earns **19,772 [20,061]**, with p5 –804 (`w02`). With costs ×1.5 and +2 s of latency it earns 14,227 [14,562]. With H1 at half size it earns 16,702 [16,591], with p5 +1,584.
   - **Plan on 10-15k per season.** Nothing here is validated on books.
7. **The ETF books lag their baskets by 7 s, but lead-lag pays only with a price cap.**
   - Correlation peaks at a 7 s lag: ARCH +0.49, REBLD +0.39 (`w01 E`).
   - With the entry capped at mid × (1 + gap/2), the lead-lag rule makes +2.7k to +3.7k over 4 rounds. Walking the book uncapped loses 6.1k to 9.7k (`w01 F`, `F2`).
   - It stays TEST-FIRST and default OFF.
8. **Feed: 5 abrupt drops and one 31.7-s outage.**
   - 22 messages were lost in season 225 and 2 in season 224, one of them official advisory #5 (`v01`).
   - Snapshots every 30.11 s carry the latest obs, a thinned track, the latest advisory, impacts and all damage reports (`v01`). They do not carry the lost intensity path.
   - The engine's snapshot handler resets the pattern posterior. Replaying each official advisory as lost and then recovered from a snapshot, 16 of 24 move a fair value past the S3 threshold. The worst is GRID +4.99, 11× the threshold (`w00 snapshot`).
   - With the two-part fix, 22 of 24 are identical to the clean feed and one more is within 0.02 of the threshold. The exception had its next snapshot 30.15 s later, a genuine gap.
9. **Testnet official advisories are fresh draws of the forecast ensemble. It does not matter to the engine.**
   - Outlook, number, age and time match in 28 of 28 advisories. `hurricane_conditions` differs with sd 0.040 (`v01`).
   - With testnet advisories, the S3 target changes in 11 of 1,551 symbol-steps, and P&L is $11,164 against $11,122 (`w00 engine`).
10. **Other teams already trade the clock.**
    - In 0-50 s of 225 r1, others net-bought 1.1-2.0 L of PORT, HAVEN, FERRY and ARCH (`v09`). The playbook's price-based crowding check never fires (`w04`).
    - Seven large market orders (GEN 558 shares, the other six a full limit) in the first 2 s of season 225 hit the ±8% walls and paid $8,522 against the 20-s mid (`w03`).
11. **Report snipes stay OFF, and the drift cannot be tested here.**

- Season 225's rounds 1-2 reports moved prices less than 1%, so the stale value per report was $0-12 in stocks and $0-53 in ETFs (`w01 I`).
- Season 225 ran against the research-mean drift. The clock-only drift trade lost 39.5k on the books and 22.7k on the consensus (`w01 H`).

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1. Feed integrity and the robust-feed design

1.1 Disconnects and losses (`v01`, `w01 A`, `w01 B`)

Drop	Round, real s	Phase	Connection lifetime	Gap to reconnect	Nearest tick (last frame vs tick)	Messages lost	First snapshot after the drop (`last_seq`)
1	224 r3, 329.4	storm	111.1 s	1.62 s	official advisory 330 (−0.56 s)	746 (official #5), 747 (storm)	31.6 s (764)
2	225 r1, 52.7	pre-storm	743.2	1.19	–	none	7.8 s
3	225 r1, 480.2	storm	426.3	1.89	track 480 (+0.19 s)	250 (storm)	2.1 s (250)
4	225 r2, 221.1	storm	699.5	**31.72**	track 220 (+1.07 s)	386-405: 16 storm, 3 wire, 1 track	49.8 s (416)
5	225 r3, 450.7	storm	1,158.5	1.32	official advisory 450 (+0.75 s)	812 (storm)	30.8 s (828)

- **The drops were abrupt** (no close frame). Every lost seq falls inside a gap (`v01`).
- **The exchange kept running during drop 4.** Twelve trades stamped inside the outage arrived after the reconnect; the first is stamped 11:20:59.80 (`w01 A`). So only the WebSocket endpoint was unreachable.
- **Drop 4's 31.7 s matches the recorder's backoff ladder** of 1, 2, 4, 8 and 16 s (`v01`).
- **All 4 storm-phase drops fell between 0.56 s before and 1.07 s after a track or advisory tick** (`w01 A`). This suggests a server-side trigger.
  - The verifier rated the feed report's claim that two connections would have saved 21.77 of 22 messages **unsupported**: it assumes the two connections drop independently.
- **Clock** (`v01`):
  - round opens 10:43:48.455, 11:00:50.037, 11:16:50.536 and 11:32:51.126;
  - residual sd 72-106 ms, worst +0.398 s;
  - the server tick runs slow, with slope 1.000395-1.000576.
  - Anchor the market clock to each storm frame rather than to a fixed 30×.

1.2 Data hygiene (`w01 C`, `v01`)

Fact	Value	Bot consequence
Largest gap between frames while connected	storm 2.12 s; pre-storm 2.54 s; aftermath 5.21 s; pause 29.57 s	Watchdogs at 3 / 8 / 8 / 35 s
Bursts	116 frames in 1 s at 225 r3 741.2 s (the final report); at most 4 news frames in 1 s	Parse off the hot path; the S1-r3 orders go out in this burst
Duplicate `trade_id`	37, every one re-sent right after a reconnect	Deduplicate by `trade_id`
Empty books	72 frames, at every close (779-781 s) and at season boundaries; 74 quotes with bid/ask `None`	Handle `None`; do not treat as a crash
Crossed or locked books	0	–
Sequence numbers	season 225 starts again at seq 0	Reset the engine and the dedup sets on a seq restart
Snapshots	123, median spacing 30.11 s; `snapshot.advisory` has no `storm`, `seq`, `round` or `type` field	See §1.3

1.3 Engine snapshot bug (`v02`, `w00 snapshot`, `w00 doublecount`)

Case	Result
Engine as is, recording replayed	The snapshot resets the pattern posterior to key (3, None) with 0 obs. The next live obs resets it again to (3, 'Omar') with 1 obs, so the recovered advisory and the snapshot track are discarded. It fired twice in 224 r3: on the join and on the recovery of advisory #5 (`v02`)
Each of the 24 season-225 official advisories lost, then recovered from the next snapshot, engine as is	16 of 24 move some fair value past k·sd + h·P. The pattern posterior moves by up to 0.986. Worst: GRID 4.99, 11.2× the threshold (r2 #2)
Fix part 1 only (add the snapshot's `storm` to its advisory, obs and track), snapshot arriving before the live message	The official label is counted twice; ARCH moves +1.30
Fix parts 1 and 2 (part 2: never feed an official advisory whose number is already known)	The double-count case is identical to the clean feed (0.000). Of the 24 recovery cases, 22 match exactly and r1 #6 is off by 0.007 (0.02 of the threshold). r3 #1 is 4.6× the threshold because its next snapshot came 30.15 s later (a genuine gap)

1.4 Robust-feed design (to build)

1. **Two market-data connections** on two network paths, such as Wi-Fi plus a hotspot, or the standby laptop. Ask for the connection limit.
   - The benefit cannot be measured from one connection: if the drops are server-side, both connections drop together.
   - Recording two connections in parallel for one season costs nothing and settles it.
2. **Merge news by (season, seq).** Forward the first copy and drop repeats. Reset the seen-set when seq restarts at 0, or when a snapshot shows `last_seq` −1 or a new date.
3. **Hold each snapshot for 1 s**, then forward it only if `last_seq` is above the highest seq already forwarded. Resync obs, track, advisory (with the §1.3 fix), impacts and damage from it. Read wire outlooks from the snapshot headlines; the feed report found each wire template maps to one outlook (`feed/wire_templates.out`, not re-verified).
4. **Market data:** for each symbol the newest server timestamp wins. Deduplicate trades by `trade_id`.
5. **Reconnect** after 0.25 s, then every 0.5 s with jitter, capped at 2 s. Do not use the 1-2-4-8-16 s ladder.
6. **Watchdogs** at 3 s in the storm, 8 s pre-storm and in the aftermath, and 35 s in pauses.
   - S3 keeps the 6-s engine-staleness rule.
   - A storm-phase gap of more than 10 s that neither connection filled switches S1-r3 to the wide ε (playbook §4.3). Drop 4 would have done so.

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2. Advisory differences and engine impact

Message type	Testnet vs research feed	Source
Storm obs, tracks, wires, impacts, damage reports, formed, dissipated	Identical. Only official advisories differ	`v01`
Official advisories (28 live: 24 in s225, 4 in s224)	Outlook, number, age and time are identical in 28 of 28. In s225 the cone differs in 24, `hurricane_conditions` in 13 and the headline in 13	`v01`
`hurricane_conditions`, s225	Differences: n = 96, mean +0.0010, sd 0.0395, max 0.160. Non-zero in 4 of 58 cases where research is 0, and in 0 of 4 where research is 1. z-sd is 1.04 for an independent 47-member ensemble draw	`v01`
Cone position, s225	Mean difference 3.8 / 7.5 / 17.8 / 40.4 units at leads of 30 / 60 / 120 / 240 s. Spreads are unbiased: 22.0 vs 21.7 … 269.6 vs 272.7	`v01`

Engine impact (`w00 engine`; bot configuration: fast params, analog library limited to seasons 226-600, fair values every 10 s at 60-540 s, 147 points):

Feed compared with research	max |Δμ| / (k·sd + h·P)	mean	Pattern max |dp|	S3 targets that differ	S3 P&L (research $11,122)
Research with the testnet advisories swapped in	≤ 0.30 (HAVEN)	≤ 0.005	0.027	11 of 1,551	$11,164
Recorded testnet feed (drops included)	up to 1.14 (AQUA, r2 s=250, inside the 31.7-s outage)	≤ 0.021	0.027	13 of 1,551	$11,065
Recorded feed, excluding the outage points	≤ 0.30	≤ 0.006	0.027	–	–
After the round-3 final report	0 for every feed	–	–	–	–

`check_recording.py` will therefore always flag official advisories. Compare them on outlook, number, age and time only.

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3. Books: spreads, depth and cost curves

3.1 The house ladder (`w01 D`; pre-storm 5-55 s, medians)

	Level 1	Level 2	Level 3	Level 4	Level 5 (wall)
Stocks: size	0.08-0.16 L	0.07-0.09 L	0.18-0.30 L	0.24 L	3.00 L
Stocks: distance from mid	8-12 bp	16-22 bp	38-49 bp	70-89 bp	797-804 bp
ETFs: size	0.39 L	0.28 L	1.20 L	0.96 L	3.00 L
ETFs: distance from mid	10 bp	20 bp	45-64 bp	91-127 bp	791-800 bp

After each open, only the wall is quoted at first. The touch first came inside 1% at a median 0.64-1.01 s after the open, and at the latest 2.50 s after (`w03`). In season 225, seven large market orders sent in that window filled 773-802 bp from the 20-s mid, at a cost of $8,522 (`w03`). They were PORT, ISLE, MAT, HAVEN, ARCH and REBLD at a full limit, and GEN at 558 shares.

3.2 Spreads and storm costs per symbol (`v03`; 0.25-s grid; cost = mean of a market buy and sell walking the book, bp of touch mid; medians unless stated)

Sym	L	Touch (sh)	Storm half-spread median / p90	Storm cost 10 sh / L/8 / L/4 / L/2 / L	Storm L/4 mean by round: 224r3 / 225r1 / r2 / r3	RT-form fit h / γ (bp, x ≤ ½)
PORT	200	20	16.8 / 56.6	18 / 23 / 41 / 67 / 328	42 / 39 / 43 / **89**	11.2 / 113
ISLE	300	29	14.3 / 25.0	15 / 19 / 35 / 55 / 310	28 / 38 / 40 / 37	10.7 / 90
AQUA	500	48	15.0 / 32.4	15 / 20 / 38 / 59 / 310	35 / 49 / 51 / 30	11.4 / 96
GRID	400	40	15.7 / 27.8	16 / 19 / 36 / 58 / 307	38 / 32 / 40 / 42	11.2 / 94
FERRY	800	80	16.8 / 38.0	17 / 21 / 40 / 64 / 304	51 / 36 / 41 / 53	12.7 / 102
RECON	500	50	15.9 / 31.9	16 / 19 / 35 / 60 / 307	31 / 34 / 42 / 53	11.0 / 96
MAT	700	70	16.1 / 30.5	16 / 19 / 36 / 60 / 306	34 / 29 / 41 / 48	11.6 / 96
GEN	600	60	16.1 / 26.5	16 / 20 / 34 / 53 / 312	33 / 31 / 37 / 43	12.7 / 80
HAVEN	250	25	23.2 / 67.8	23 / 28 / 51 / 80 / 326	60 / 42 / 59 / **94**	16.0 / 129
REIN	200	19	12.2 / 15.6	12 / 16 / 29 / 47 / 316	32 / 28 / 27 / 29	7.8 / 78
ARCH	50	20	15.8 / 35.6	16 / 16 / 16 / 20 / 35	15 / 15 / 19 / 27	14.0 / 11
REBLD	100	40	15.5 / 29.6	16 / 16 / 16 / 19 / 32	14 / 14 / 17 / 24	14.5 / 7

3.3 Costs by phase (`v03`; L/4 clip, median bp)

Sym	pre 3-60 s	H1 entry 50-58 s	72-80 s	80-120 s	storm 120-540 s	after 540-780 s
PORT	22	22	64	169	41	23

Sym	pre 3-60 s	H1 entry 50-58 s	72-80 s	80-120 s	storm 120-540 s	after 540-780 s
ISLE	22	24	86	76	35	25
AQUA	24	24	85	76	38	28
GRID	22	22	79	115	36	31
FERRY	20	21	76	66	40	31
RECON	22	21	57	108	35	32
MAT	25	24	56	86	36	31
GEN	26	26	91	86	34	27
HAVEN	21	21	141	254	51	28
REIN	25	26	30	27	29	29
ARCH	10	10	0.2	46	16	10
REBLD	10	10	28	32	16	10

- **Phase effects.** The 72-120 s window costs 2-7× the storm. Pre-storm and aftermath clips are cheap.
  - ARCH's 0.2 bp at 72-80 s is not a real cost. At that point the ARCH book is clamped one tick inside its stale levels (see §4.5).
- **The table means are skewed by one round.** PORT's storm L/4 mean is 54 and HAVEN's 64, because of 225 r3 (89 and 94). The medians are 41 and 51 (`v03`).
- **Follow-on** (`w01 G`). After another taker's order of at least 0.2 L, an L/4 clip on the same side costs more than it did just before:
  - stocks: median 27 bp just before, 52 bp at +2 s (n = 100);
  - ETFs: 16 bp just before, 45 bp at +2 s (n = 210).
  - These are gross figures: they include whatever the price did after the order.
  - The re-backtest's follow-on term of 100 bp per L of our own same-side flow in 8 s for stocks (about 25 bp per L/4 clip) is in line with this. Its 48 bp per L for ETFs looks low against this check.
- **What the bot should use:** piecewise-linear interpolation of the phase `table` in `books/cost_model.json`, plus the follow-on term, plus the impact kernel. The impact kernel is fitted on other participants' flow (`rebacktest/r00_followon.out`) and was not re-measured independently. Treat clips above about 0.7 L as off-limits for stocks.

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4. Which edges exist on real books

Edge	On the books?	Size on books vs consensus (same rounds)	Verdict
H1 formation leg	Yes, late. Sign agrees in 29 of 33; size ratio 1.20 (`v04`). Book half-move at a median 72.0 s for stocks (range 66.0-89.6 s) and 77.3 s for ETFs (`w06`)	Full size, exit 72-78 s: **+65** vs +7,389. Half size: –1,658 vs +4,401 (`v04`)	TEST-FIRST, then GO with **no exit before 80 s** (S3 inherits the H1 book)
In-storm drift (clock-only)	Sign agrees in 30 of 33 (`w01 H`), but season 225 ran against the research mean	Drift trade **–39,476** vs –22,707 consensus RT (`w01 H`)	Fallback only. No evidence either way
Report windows, rounds 1-2 (S2, S1)	The book is stale for 6-8 s (`v05`)	Stale value per report in 225 r1-r2: $0-12 in stocks, $0-53 in ETFs. 225 r3 preliminary: $263-352 in stocks at +1-2 s (`w01 I`)	OFF until the latency test (§7)
S1-r3	Yes. The book is stale 6-8 s after the final report	A sweeper from +1 s finds at least **$924 / $927** per round 3. The consensus oracle is $1,335 / $1,407 at 742 s and $196 / $225 at 744 s (`v05`)	**GO; keep the taker**; sanity check at ≥ 756 s
ETF lead-lag	Yes, a 7-s lag (`w01 E`)	**+2,673 to +3,684** over 4 rounds with a capped entry; **–6,145 to –9,675** walking the book (`w01 F`, `F2`)	TEST-FIRST, default OFF; capped entries only

Edge	On the books?	Size on books vs consensus (same rounds)	Verdict
S3 engine taker	The signal holds against the 600-s level (`v08`)	S3 alone in season 225: **–3,790 / –2,175** on books vs +3,569 / +3,000 on consensus. Round 3: +6,763 / +7,738 on books (`v07` , `v12`)	GO (core), but score it on fills next testnet season

4.1 Formation and H1 (`v04` , `v09` , `w04` , `w06`)

H1 exit window, full size, entry cap kept	Books r1 / r2 / r3 = total	Consensus RT, capped
72-78 s (playbook)	+2,947 / +872 / –3,754 = **+65**	+7,389
80-86 s	+808	+8,168
84-90 s	+2,785	+12,971
86-92 s	+4,551	+17,026
120-126 s	+24,479	+29,803

- **The 72-78 s exit sells into a stale book.** In r1 at 74 s, GRID, FERRY and ARCH had moved +0.15%, +0.16% and +0.26% on the book, against +5.0%, +10.0% and +4.0% on the consensus. They caught up by 80 s (`v04`).
- **Later exits look better only because of the drift.** Season 225 drifted the H1 way, so later exits are flattered by drift, not by the formation jump (`v04`).
- **The entry cap (48-s mid ± 0.5%) binds, and it protects.** In r2, AQUA filled 173 of 500, GRID 188 of 400 and GEN 261 of 600. In r3, PORT filled 86 of 200 and RECON 85 of 500 (`v04`).
  - The books report replayed H1 without the cap and got –2.7k. The verifier attributes the difference to the missing cap.
- **Crowding is real.**
  - In 0-50 s of r1, others net-bought PORT +1.52 L, HAVEN +1.96 L, FERRY +1.60 L and ARCH +1.06 L, and sold REBLD –0.99 L. In r3 they bought HAVEN +1.94 L and ARCH +2.06 L (`v09`).
  - In 60-80 s of r3, others sold ARCH –4.62 L into the stale book (`v09`).
  - The playbook's price-based crowding skip never fires: the 30→48 s book move was at most 0.21% (`w04`).
  - r1, a crowded round, was H1's best round on the books. So there is no evidence the skip helps.
- **This season's formation-jump direction varied.** The book moved HAVEN +21.8%, +6.6% and –10.8% from 58 to 90 s in r1, r2 and r3 (`w04`).

4.2 Reports and S1-r3 (`v05` , `w01 I`)

- **The book reprices 3-4 s after the consensus.** For the 7 stock moves of at least 1%, the median share of the move done is:

+2 s	+4 s	+6 s	+8 s	+10 s
0.07	0.20	0.36	0.91	1.00

  The consensus has done 0.99 by row +2 (+4 s). The books report's "+1.0 s / 0.0 s" lag was corrected by the verifier.

S1-r3	224 r3	225 r3
Final report received	740.26 s	740.23 s
Sweeper from te + 0.3 / 1 / 2 / 4 / 6 s (lower bound; ignores re-posts)	$937 / 924 / 679 / 619 / 362	$966 / 927 / 834 / 545 / 229
Value range including re-posted stale quotes (verifier's range)	$937-1,250	$966-1,386
Value other participants took beyond S ± ε	$934	$2,307
Consensus oracle at 742 / 744 s, 25 bp	$1,335 / $196	$1,407 / $225
Sanity check |mid – S| > max(3ε, 0.5%) would cancel at 746 / 750 / 752 / 756 s	PORT, HAVEN, ARCH / ARCH / ARCH / none	PORT, HAVEN, ARCH / HAVEN / none / none

- The re-backtest priced the S1-r3 taker at −$15 because it fills on the consensus at 744 s, after the consensus has repriced. Its `REC without S1−r3` is 19,786 against 19,772 with it (`w02`). That fill model does not match the books, so **its advice to drop the taker is unsupported**.

4.3 S3 on the books (`v07`, `v08`, `v12`)

Season 225, S3 only	r1	r2	r3	Total
Independent replay, consensus (`v12`)	+1,949	−4,448	+5,499	+3,000
Independent replay, books (`v12`)	−505	−9,407	+7,738	**−2,175**
`r11` (REOPT without H1), consensus	+2,182	−4,680	+6,068	+3,569
`r11`, recorded books, 0.3 s	−779	−9,773	+6,763	**−3,790**

- **The S3 signal holds against the post-storm book level.** Correlation of (engine − book) with (book at 600 s − book) is 0.94, 0.74, 0.13 and 0.50 in 224 r3 and 225 r1 / r2 / r3.
  - The engine's RMSE against the 600-s book level is 137, 181, 228 and 330 bp; the book's own is 344, 241, 148 and 306 bp. So the engine beats the book in 224 r3 and 225 r1 and loses in 225 r2 (`v08`).
- **In round 3, the sign rule earns about the same on books as on the consensus.** The rule (|engine − price| > 75 bp, scored against settlement) earns +304 bp per signal on book prices (n = 419), against +338 bp on the consensus (`v08`).
- **Flattening after the storm-end snap barely helps.** In r1 the HAVEN book snapped from 56.2 to 52.75 at 546-552 s (`v10`). Flattening rounds 1-2 over 552-568 s instead gives books r1 +63 and r2 −9,453 (`v12`).

4.4 Drift (`w01 H`)

- Season 225's in-storm move from 74 to 538 s ran against the research mean. HAVEN's book rose +26.3%, +30.1% and +7.7% by round; its consensus rose +7.1%, +22.1% and +6.2%.
- The clock-only drift trade (playbook §4.7, worked L/4 clips) lost:

	r1	r2	r3	Total
Books	−13,633	−18,275	−7,568	−39,476
Consensus RT	−4,370	−14,986	−3,351	−22,707

- This is one season against a mean measured over 600, so it neither confirms nor refutes the drift.

4.5 ETF lag (`w01 E`, `w01 F`, `w01 F2`, `v03`)

	ARCH	REBLD
Correlation of 2-s changes, ETF(t) vs basket NAV(t − k)	+0.08 at 0 s, peak **+0.49 at 7 s**	+0.10 at 0 s, peak **+0.39 at 7 s**
RMS of log ETF(t)/NAV(t − k), demeaned per round	60 bp at 0 s, **32 bp at 7 s**	42 bp at 0 s, **28 bp at 7 s**
Touch through NAV by more than 50 bp	7.9% of 60-776 s; 65 episodes, median 1.5 s	5.3%; 46 episodes, median 1.5 s
68-80 s after formation: NAV − ETF, 225 r1 / r2 / r3	+0.7 to +6.5% / −0.6 to +2.7% / −2.4 to −0.4%	−2.3 to −0.2% / −0.8 to +0.9% / −0.7 to +1.3%
Size priced better than NAV ∓ 0.3% at 68-80 s, median, % of limit	216 / 216 / 158	216 / 28 / 0
Storm half-spread, median / p90 (`v03`)	15.8 / 35.6 bp	15.5 / 29.6 bp

The playbook §4.5 lead-lag rule replayed on the books, both ETFs, full limit, 4 rounds (224 r3 from 222 s). Our fills do not deplete the book, which is optimistic:

Entry order	Decision grid	0.3 s latency	1 s	2 s
Walk the book (no cap)	0.5 s	−6,145	−7,824	−9,675

Entry order	Decision grid	0.3 s latency	1 s	2 s
Marketable limit capped at mid × (1 + gap/2)	1 s	+2,673	+3,016	+3,230
Same, capped	0.5 s	+3,684	+3,483	+2,382

- **The book lag is real, but the cap does the work.** Without it the rule pays the spread on entries the book has already corrected.
- **Do not swap the ETF legs for lead-lag.** On the consensus, the ETF legs inside S3 and H1 are worth 19,772 with ETFs against 15,902 without [20,061 vs 16,256] (`w02`). Lead-lag's book value on one season is no larger than that.
- **A formation-only ETF rule would conflict with H1's ARCH leg.** ARCH is stale at 68-80 s, but H1 already holds the ARCH limit.

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5. Re-backtested expected value with measured costs

On the consensus. Holdout 401-600, train 1-400 in [brackets], $ per season at full limits. Fills use the `cost_model.json` phase tables, follow-on and impact; the report race is lost.

"REC" is the re-backtest's recommended configuration: REOPT with H1 at full size, entered in L/4 clips at 50-56 s. This document keeps H1 at ½ size in round 1 until the round-1 check passes (§6), so the "H1 at half size" row is the floor for round 1.

Configuration	Mean	p5	Seasons > 0	Source
Plan as written, RT fill model (playbook)	20,346 [19,504]	2,313	96%	playbook, `r01`
Plan as written, measured fills	**10,713 [9,811]**	−7,821	83% [81%]	`v06`
REOPT (H1 in L/8 clips, 42-56 s)	19,778 [20,145]	−663 [1,411]	94%	`v06`
REC	**19,772 [20,061]**	−804 [1,323]	94% [96%]	`v06`, `w02`
REC under the RT fill model	24,338 [24,504]	3,728	96%	`w02`
REC, costs ×1.5	14,993 [15,358]	−4,588	89%	`w02`
REC, costs ×2	10,653 [11,119]	−8,344	78%	`w02`
REC, +2 s latency	18,991 [19,231]	−1,526	94%	`w02`
REC, +2 s and costs ×1.5	**14,227 [14,562]**	−5,939	87%	`w02`
REC, 60-120 s costs ×3	15,910 [16,550]	−4,341	86%	`w02`
REC, H1 at half size	16,702 [16,591]	1,584 [155]	96%	`w02`
REC, H1 without RECON, MAT, GEN, REBLD	18,660 [18,755]	1,015 [2,526]	95%	`w02`
REC without H1 (S3 + S1-r3)	12,453 [12,024]	−2,972	91%	`w02`
REC without ETFs	15,902 [16,256]	−1,556	94%	`w02`
REC without S1-r3	19,786 [20,071]	−768	94%	`w02`

REC uses 1,229 actions per season and peaks at the 150-per-minute cap (`w02`).

On the recorded books (season 225 only; `r11`, reproduced exactly in `v07`):

Configuration	Consensus	Recorded books, 0.3 s	Books, +2 s
REOPT	34,872	23,210	23,569
REOPT, forced H1 exit 72-86 s	24,638	11,360	11,540
Plan	−4,991	−16,323	−16,062
S3 only	3,569	−3,790	−3,556

- **REOPT's +23.2k on the books is not a validation.** H1 inventory held through a season that drifted the H1 way made +48.4k, and S3 lost 25.4k (`v07`).
- **Planning number: 10-15k per season.**
  - The top of the range is REC at costs ×1.5 with +2 s of latency (14.2k). The bottom is REC at costs ×2 (10.7k).
  - Both are consensus-priced. The verifier rates the planning number unsupported on books.

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6. Changes to the playbook strategies and bot parameters

Item	Playbook before	Now [testnet]	Evidence
Fill model h	0.25%	Use the phase tables (`cost_model.json`). Storm intercept 11-16 bp for stocks, 14 bp for ETFs	`v03`
Fill model γ	0.5% per full-limit clip	Storm 0.8-1.3% for stocks (x ≤ ½), 0.07-0.11% for ETFs. 80-120 s costs 2-7× the storm. A 3 L wall beyond about 0.7 L	`v03`, `w01 D`
S3 clip	¼ L per 2 s	**L/12 per 2 s** (bot units below)	`v06`, `w05`
S3 entry threshold	k·sd + γ·mid (75 bp)	k·sd + 2.0 × (walk cost of one L/12 clip + 36 bp; 17 bp for ETFs). That is 107-138 bp for stocks in the storm and 69-73 bp for ETFs; 138-342 bp at 80-120 s. k stays 0.1 in rounds 1-2 and 0.2 in round 3	`w05`, `v06`
S3 order cap	min(μ − thr, ask × 1.005)	Same, and **never beyond mid ± 1.5%** (the wall sits about 8% out)	`w01 D` [I]
Flatten, rounds 1-2	562-568 s, ¼ clips	**540-568 s, L/12 clips**	`v06`
H1 entry	50-56 s, L/4 clips, capped at the 48-s mid ± 0.5%	Unchanged. The cap binds and protects	`v04`
H1 exit	72-78 s	**No forced exit; S3 inherits the H1 book at 80 s.** If the engine is not live by 76 s, exit over 84-98 s in L/8 clips [I]	`v04`, `w06`, `v06`
H1 size	Round 1 ½; rounds 2-3 full if the round-1 check passes	Unchanged. On the consensus, full size from round 1 earns 19,772 against 16,702 at half size, but its p5 is −804 against +1,584. On the books H1 is unproven	`w02`, `v04`
H1 crowding skip	30→48 s price move > ⅓ of the jump	**Drop it.** It never fires on the books	`w04`, `v09`
H1 round-1 check	Cost ≤ ½ of the 58→74 s jump	**Use the 58→90 s book move** of HAVEN, PORT and ARCH against our own entry cost per share (48-s mid to fill); pass if cost ≤ ⅓ of the move [I]	`v04` (the book has barely moved by 74 s)
H1 rebuilder legs	In	Keep at base cost. Drop RECON, MAT, GEN and REBLD from H1 if round-1 costs are ≥ 1.5× the tables (p5 then 1,015)	`w02`
S1-r3	Take at 740 s, top up every 2 s; sanity check at 746 s	**Keep the taker.** Sweep every 0.5-1 s until 750 s. **Sanity check at ≥ 756 s**	`v05`
ETF lead-lag	TEST-FIRST	**TEST-FIRST, default OFF.** If enabled: entries only as marketable limits capped at mid × (1 + gap/2), never a book walk. Keep the ETF legs in S3 and H1	`w01 F`, `F2`, `w02`
Report snipes (S2; S1 in rounds 1-2)	OFF	OFF. The book is stale 6-8 s, but latency is unmeasured	`v05`, `w01 I`
Round opens	–	**No marketable orders in the first 3 s of a round**	`w03`
Feed	One connection; 6-s and 10-s rules	Two connections, merge and dedup, fast reconnect, phase watchdogs, the engine snapshot fix (§1.3-1.4)	`v01`, `v02`, `w00`
Fill sanity	3 clips with slippage > 3 × (h + γ/4)	3 clips with slippage > 3 × the phase-table cost	`v03`
Action budget	150 per minute	Unchanged. REC uses 1,229 actions per season and reaches 150 in a minute	`w02`
Planning EV	20.3k	10-15k, consensus-priced	§5

Bot units (`w05`; exit band = phase mean half-spread):

Sym	L	S3 clip L/12 (sh)	H1 clip L/4	L/8 (H1 fallback exit)	Entry cost term, storm / 80-120 s / after (bp)	Exit band, storm / 80-120 s (bp)
PORT	200	16	50	25	123 / 315 / 101	24 / 114
ISLE	300	25	75	37	107 / 165 / 99	16 / 43
AQUA	500	41	125	62	112 / 170 / 98	18 / 47
GRID	400	33	100	50	107 / 205 / 99	17 / 63
FERRY	800	66	200	100	117 / 138 / 102	21 / 32
RECON	500	41	125	62	110 / 195 / 99	18 / 57
MAT	700	58	175	87	110 / 164 / 100	18 / 43
GEN	600	50	150	75	109 / 161 / 100	18 / 42
HAVEN	250	20	62	31	138 / 342 / 104	31 / 123
ARCH	50	4	12	6	73 / 129 / 58	19 / 46
REBLD	100	8	25	12	69 / 101 / 55	17 / 33

REIN stays out of H1 and S3.

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7. What still needs testing

#	Open question	Why it matters	How
1	Our own fills and impact	Every cost above comes from other participants' orders	The 1-lot plan below, then a full testnet season at 25% size
2	S3 on books	The only book evidence (one season) is negative in rounds 1-2	Full testnet season at 25% size, with S3 P&L scored on fills. If S3 is negative on fills in rounds 1-2 again, restrict it to round 3 [l]
3	Latency: message → order → ack → visible fill	The S2/S1 gate (p95 < 1.5 s); S1-r3 sweep timing; the +2 s assumption	Test T4-T5
4	Price protection and order semantics	Sets the mid ± cap; whether remainders rest; IOC	Tests T2-T3 and Slack
5	Whether the drops are server-side	Decides whether a second connection helps	Record two connections in parallel for one season (zero risk)
6	Engine fix and merged feed	The §1.3 bug is material	Replay this recording through the patched engine and the merge layer; expect 22 of 24 recovery cases exact and r3 #1 off because of its 30-s gap
7	Sample size	3 formations and 8 reports	Record at least 2 more seasons (books plus news) and re-run `verify/` and `writer/`
8	Testnet house = live house?	The book levels differ from `market.csv`; depth and walls	Slack

Proposed 1-lot order test plan (needs team approval before it runs; nothing has been sent)

Scope and limits.

- Testnet host `cornelltradingcompetition.org` only, with the team bot key. Never on the live exchange.
- 1 share per order (1 unit for create/redeem).
- At most 60 order actions per round and 10 per 10 s.
- Kill switch armed. Every test position is closed with a 1-lot order in the same round.
- **Stop rules:**
  - an order state unknown for more than 3 s;
  - any reject code not already listed;
  - cumulative realised cost above $100.

- Operator: A. Watcher: D. The team approves in writing before T1.

Test	When	Orders	Measures	Decides
T1. Order lifecycle: 1-lot buy at touch –2% (rests), `GET /orders/open`, `DELETE /orders/{id}`, cancel-all by symbol	pre-storm 10-40 s	6	Ack and cancel latency; response schemas	Parsers
T2. Marketable round trip: 1-lot buy at the ask, 1-lot sell at the bid, on PORT, HAVEN and ARCH	aftermath 600-700 s	12	Fill price vs touch; time to `/positions`; whether our trade prints on the trade channel	Fill model at 1 lot
T3. Price protection: 1-lot limits 1%, 3% and 10% through the touch, plus one 1-lot market order, on PORT	aftermath 600-720 s	8	Reject, cap, or fill price	The mid ± cap
T4. Latency ladder: one 1-lot marketable order every 30 s in 90-540 s, alternating sides	storm	32	News → send → ack → `/positions`, p50 / p95	The +2 s assumption
T5. Burst latency: 1-lot PORT at te + 0.3 s after the 570-s and 740-s reports, closed 20 s later	reports	8	Ack p95 under a 116-frames-per-second burst	S2/S1 gate; S1-r3 sweep
T6. Formation fills: 1-lot PORT buy at 52 s (capped at 48-s mid + 0.5%), sold at 84 s. In another round, 1-lot ARCH when NAV – ETF > 0.5% at 68-80 s, capped at mid × (1 + gap/2), sold at 84 s	50-90 s	4 per round	Whether stale levels fill at our price	H1 exit timing; the ETF cap
T7. Create/redeem 1 unit (if offered); self-trade (rest a 1-lot buy, send a 1-lot marketable sell at it); limit-room cancel	aftermath	≤ 6	P&L at marks; rejects; cancels	Slack Q4 in the playbook

8. New Slack questions

These say nothing about the house's reaction timing or its formation behaviour.

1. Our market-data WebSocket dropped five times in an hour with no close frame (connection lifetimes 111-1,159 s). Is there a server-side idle or heartbeat timeout, or a recommended ping interval? Will the live venue use the same gateway?
2. How many concurrent market-data connections may a team or key hold? Can we keep two open for redundancy?
3. Is there a REST endpoint for the current snapshot, or for news since seq N, so a client can resync after a drop?
4. Is the snapshot schema final? Its `advisory` has no `storm`, `seq`, `round` or `type` field. Will live snapshots also come about every 30 s?
5. On the testnet, official advisories differ from the published research feed in `hurricane_conditions`, cone and headline, while storm observations, tracks, wires and reports are identical. Are official advisories regenerated on each run, and will live advisories come from the same generator?
6. Is `trade_id` globally unique and safe to deduplicate on? The last trade per symbol is re-sent after a reconnect. Is there an aggressor-side field or a private fills channel?
7. Market-data timestamps are local wall clock without a time zone, and news `time` is the market clock. Will live use the same conventions? Is there a server-time endpoint with millisecond precision?
8. At each close every book goes empty and quotes carry a null bid and ask. Are resting orders cancelled at the close?
9. Is the testnet house the same pricing model and configuration as the live house: starting prices, spreads, depth and order sizes?
10. Is there an opening auction or restricted period in the first seconds of a round? What happens to a limit order priced far through the touch: is it filled at the touch, capped, or rejected?
11. Is the tick size 0.01 for every symbol, ETFs included?

Appendix: reproduction

Output	Script	Content
`verify/v01_feed.out`	`v01_feed.py`	Clock fits, connections, lost and different messages, advisory statistics, snapshot keys

Output	Script	Content
`verify/v02_snapshot_bug.out`	`v02_snapshot_bug.py`	Engine reset on snapshot advisories
`verify/v03_costs.out`	`v03_costs.py`	Half-spreads, costs by phase and round, RT-form fits
`verify/v04_h1.out`	`v04_h1.py`	Formation jump, H1 replay on books vs consensus
`verify/v05_reports.out`	`v05_reports.py`	Report reprice timing, S1-r3 sweeper, sanity-check timing
`verify/v06_rerun_*.out`	`v06_rerun.py`	Re-backtest reruns
`verify/v07_r11_repro.out`	`rebacktest/r11_book_replay.py`	Season-225 book replay
`verify/v08_book_vs_truth.out`	`v08_book_vs_truth.py`	Engine, book and consensus accuracy; S3 signal
`verify/v09_flow_open.out`	`v09_flow_open.py`	Others' net flow; season-225 open vs packet
`verify/v10_gap_path.out`	`v10_gap_path.py`	Book vs consensus by time shift; storm-end paths
`verify/v12_s3_book.out`	`v12_s3_book.py`	Independent S3 replay on books
`writer/w00_engine_sens_fast.out`	`feed/engine_sens.py` (re-run)	Engine sensitivity to testnet advisories and gaps
`writer/w00_snapshot_resync.out`	`feed/snapshot_resync_check.py` (re-run)	Snapshot recovery, engine as is vs patched
`writer/w00_doublecount.out`	`feed/doublecount_check.py` (re-run)	Fix part 1 vs parts 1+2
`writer/w01_checks.out`	`writer/w01_checks.py`	A drops; B snapshot recovery; C hygiene; D ladder; E ETF lag; F/F2 ETF lead-lag replay; G follow-on; H drift; I report stale value
`writer/w02_sens_holdout.out`, `w02_sens_train.out`	`writer/w02_sens.py`	REC and its sensitivities and marginals
`writer/w03_open.out`	`writer/w03_open.py`	Book and large trades right after each open
`writer/w04_crowding.out`	`writer/w04_crowding.py`	Playbook crowding check on the books
`writer/w05_params.out`	`writer/w05_params.py`	Bot-unit clips, thresholds and exit bands from `cost_model.json`
`writer/w06_t50.out`	`writer/w06_t50.py`	Formation half-move times, summarised from `v04`